Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/15/2004
PI Name:	Wilcox, William R Ph.D.		
Project Title:	Residual Gas Effects on Detached Solidification in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Electronic materials		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	13699-5705	**Congressional District:**	23
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	06/01/2000	**End Date:**	11/28/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	2
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Regel, Liya (Clarkson University)		
Grant/Contract No.:	NAG8-1703		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	A system was perfected that controls the oxygen concentration over a sessile drop of molten semiconductor InSb while permitting the surface on which the drop rests to be tilted. The drop shape is measured while the surface is horizontal in order to give the surface tension of the melt in addition to the static contact angle. Contact angle hysteresis is measured by slowly tilting the surface until the drop begins to move. Our long-term goal is to make detached solidification reproducible, which requires a full understanding of the mechanisms underlying it. Our Moving Meniscus Model of steady-state detachment predicts that it depends strongly on the surface tension of the melt and the advancing contact angle with the ampoule wall. The objective of the current task is to determine the influence of residual gases on the surface tension and contact angle of molten semiconductors on typical ampoule materials. Our initial focus is on the influence of oxygen on indium antimonide (InSb) on clean silica (“quartz”). Greatly superior crystals are produced when detached solidification is achieved, with orders of magnitude fewer dislocations, grain boundaries and twins. Some behavior observed during detached solidification has been attributed to the presence of residual gases in the ampoule,		

	particularly oxygen and water vapor. However, there are no data on the influence of these gases on the surface tension or contact angle of molten semiconductors, except some excellent Japanese results for the influence of oxygen on the surface tension of molten silicon.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We now understand how to produce superior crystals by detached solidification not only in space but also on Earth. Furthermore, we also now understand the interaction between oxygen, other gases, and the semiconductor melt. This is helpful in growing semiconductors by a variety of techniques.
Task Progress:	Sessile drop experiments were performed on molten InSb containing ~0.1%Ga on clean quartz (fused silica) surfaces. A cell was constructed through which a mixture of argon, oxygen, and sometimes hydrogen were flowed. When the oxygen partial pressure exceeded approximately 10-6 bar in the absence of hydrogen, oxide formed on the surface of the melt that prevented it from deforming or moving on the substrate. This oxide was highly enriched in Ga. Reduction of the oxygen pressure or addition of 10% hydrogen to the argon caused the oxide to dissociate so that a shiny drop formed that easily moved across the substrate when the apparatus was tapped or the substrate was tilted slightly. Experiments performed on such shiny drops revealed that the surface tension of the molten InSb decreased with increasing hydrogen and oxygen partial pressures, and was relatively independent of temperature. It was estimated that this oxygen influence was caused by adsorption of ¼ of a monolayer of oxygen on the melt surface. The contact angle on quartz decreased with increasing oxygen pressure, and was relatively independent of hydrogen pressure or temperature, indicating that the solid-melt surface energy also was decreased by adsorbed oxygen. Both surface tension and contact angle appeared to vary with the square root of the oxygen partial pressure, indicating that oxygen dissociates when it dissolves in or adsorbs on the InSb. Raising the temperature caused a momentary decrease in oxygen concentration in the exit gas. This was attributed to adsorption of additional oxygen onto the surface of the melt. Tapping the apparatus increased the contact angle by about 0.6 degrees, presumably because tapping overcame contact line sticking. Contact line sticking also caused a few degrees difference in upper and lower contact angles when the substrate was tilted until the drop rolled down. Recently we have been able to reduce the oxygen partial pressure to below 0.1 ppm at about 1 atm. We have obtained gallium-free InSb and are using this for experiments. Thus far, the results do not seem substantially different from those obtained with Ga-doped InSb.
Bibliography Type:	Description: (Last Updated: 03/03/2005)
Abstracts for Journals and Proceedings	Kota, A.K.; Wilcox, W.R.; Regel, L.L. "Influence of oxygen on the contact angle and surface tension of molten indium antimonide" Annual Technical Meeting, New York State Center for Advanced Materials Processing, Saratoga Springs none , May-2003
Articles in Peer-reviewed Journals	Wang, J.; Regel, L.L.; Wilcox, W.R. "Detached solidification of InSb on Earth, J. Crystal Growth 260, 590-599 (2004)." PACS , Jan-2004
Dissertations and Theses	Kota, A.K. "Influence of Oxygen on the Contact Angle and Surface Tension of Molten Indium Antimonide" M.S. Thesis, Clarkson University Aug-2003
Dissertations and Theses	Ramakrishnan, S. "Development of a Technique for the Determination of Influence of Oxygen on the Contact Angle and Surface Tension of Indium Antimonide" M.S. Thesis, Clarkson University Jul-2003
Papers from Meeting Proceedings	Regel, L.L.; Wilcox, W.R.; Ramakrishnan, S.: Kota, A. "Improved Crystal Quality by Detached Solidification in Microgravity" Proceedings of the NASA 2002 Microgravity Materials Science Conference, NASA/CP -2003-212339, edited by D. Gillies, N. Ramachandran, K. Murphy, D. McCauley, and N. Bennett, pp 688-694 Feb-2003